

Ferdous Hasan

Computer Science graduate specializing in Artificial Intelligence and Machine Learning (AI/ML) and full-stack development.

☎ +919883160649 ✉ ferdoushasan.dev@gmail.com 🔗 LinkedIn 🐙 GitHub 📍 Kolkata, India

PROFESSIONAL SUMMARY

Experienced in building scalable web applications, integrating machine learning models, and deploying cloud-native solutions. Strong focus on real-time systems, backend engineering, and performance optimization.

EDUCATIONS

Techno India University | Kolkata, WB

2022 - 2026

B.Tech in Computer Science & Engineering

- Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Machine Learning.

PROJECTS

AI Face Attendance System (Computer Vision) | 🔗

- Designed** and **implemented** a real-time face recognition **computer vision pipeline** utilizing **FaceNet** and **OpenCV**, achieving **98% accuracy**.
- Developed** high-performance **RESTful APIs** using **FastAPI** and **SQLAlchemy** to enable efficient real-time attendance tracking.
- Deployed** a containerized system on **Oracle Cloud** with **Docker**, ensuring **high availability**, **fault tolerance**, and **scalability**.

Technologies / Tools Used : FaceNet, OpenCV, FastAPI, SQLAlchemy, Docker, Oracle Cloud

Luna AI Assistant | 🔗

- Developed a high-performance, cross-platform desktop AI assistant using **React/Tauri** for the frontend and **FastAPI** for the backend logic.
- Engineered a real-time voice interaction pipeline featuring Voice Activity Detection (Silero VAD), local Speech-to-Text (faster-whisper), and Text-to-Speech (Piper).
- Integrated native OS-level controls allowing the AI to interact with system applications across macOS and Windows.
- Implemented complex file processing capabilities, including real-time PDF parsing, computer vision, and local persistent memory for contextual conversations.

Technologies / Tools Used : React, TypeScript, Tauri, Python, FastAPI, WebSockets

VeritasAI - Deepfake & Synthetic Media Detection Platform | 🔗

- Architected a full-stack enterprise application to detect AI-generated media using a custom hybrid machine learning pipeline.
- Engineered a multi-heuristic backend with FastAPI, integrating PyTorch Vision Transformers for temporal frame classification and OpenCV (FFT) for structural anomaly detection.
- Developed a responsive, low-latency frontend using React, Vite, Tailwind CSS, and Framer Motion, featuring drag-and-drop file streaming and real-time visualization.

Technologies / Tools Used : React, Vite, Tailwind CSS, FastAPI, PyTorch, HuggingFace, OpenCV, NumPy.

Nexora - Real-Time Video Conferencing Platform | 🔗

- Designed and developed a scalable real-time communication system using Next.js and Stream SDK supporting 100+ concurrent users with low-latency streaming (<200ms).
- Achieved **25% reduction in latency** through optimized state management and server-side rendering.
- Implemented secure authentication and role-based access control using Clerk.

Technologies / Tools Used : Next.js, Stream SDK

SKILLS

Databases :	MySQL, PostgreSQL, SQLAlchemy, Schema Design
Frameworks & Libraries :	TensorFlow, OpenCV, FaceNet, Scikit-learn, NumPy, Pandas, React.js, Next.js, FastAPI, Flask, Node.js, REST APIs, Tailwind CSS, Redux
Languages :	English, Hindi
Programming Languages :	Python, Java, C++, SQL, HTML/CSS, JavaScript, TypeScript, C
Soft Skills :	Problem Solving, Self-Directed Learning, Attention to Detail, Technical Communication, Team Collaboration, Adaptability
Tools & Platforms :	Oracle Cloud (OCI), Docker, Git/GitHub, Linux, CI/CD, Postman

CERTIFICATIONS

NVIDIA DLI: AI for All – Generative AI and Deep Learning Fundamentals.

Eduskills: Android Developer Certification (Java/Kotlin, SDK).